Data: 24/09/2026
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Roll no: 255hsbb011

Chip-seq analysis:

Aligning to ref genome and sorting it

1. Chip Sample C

Bash

```
bowtie2 -p 8 -x E_coli_BW25113_index -U SO_4933_C_CHR1_R1.fastq.gz | samtools
sort -@ 4 -o C_CHR1_sorted.bam -
samtools index C_CHR1_sorted.bam
```



2. Input Control C (INR)

Bash

```
bowtie2 -p 8 -x E_coli_BW25113_index -U SO_4933_C_INR1_R1.fastq.gz | samtools
sort -@ 4 -o C_INR1_sorted.bam -
samtools index C_INR1_sorted.bam
```



3. Chip Sample D

Bash

```
bowtie2 -p 8 -x E_coli_BW25113_index -U SO_4933_D_CHR1_R1.fastq.gz | samtools
sort -@ 4 -o D_CHR1_sorted.bam -
samtools index D_CHR1_sorted.bam
```

```
(fastq_files_24_09) (fastq_files_24_09) ibab@IBAB-MSc-Comp11:~/bio-env/fastq_files_24_09$ bowtie2 -p 8 -x E_coli_BW25113_index -U SO_4933_D_CHR1_R1.fast
q.gz | samtools sort -@ 4 -o D_CHR1_sorted.bam -
samtools index D_CHR1_sorted.bam
[WARNING] Failed to launch x86-64-v3 version, staying with default
[WARNING] Failed to launch x86-64-v3 version, staying with default
17485990 reads; of these:
  17485990 (100.00%) were unpaired; of these:
    1475353 (8.44%) aligned 0 times
    15537443 (88.86%) aligned exactly 1 time
    473194 (2.71%) aligned >1 times
91.56% overall alignment rate
[bam_sort_core] merging from 4 files and 4 in-memory blocks...
```

4. Input Control D (INR)

Bash

```
bowtie2 -p 8 -x E_coli_BW25113_index -U SO_4933_D_INR1_R1.fastq.gz | samtools
sort -@ 4 -o D_INR1_sorted.bam -
samtools index D_INR1_sorted.bam
```

```
[bam_sort_core] merging from 4 files and 4 in-memory blocks...
(fastq_files_24_09) (fastq_files_24_09) ibab@IBAB-MSc-Comp11:~/bio-env/fastq_files_24_09$ bowtie2 -p 8 -x E_coli_BW25113_index -U SO_4933_D_INR1_R1.fast
q.gz | samtools sort -@ 4 -o D_INR1_sorted.bam -
samtools index D_INR1_sorted.bam
[WARNING] Failed to launch x86-64-v3 version, staying with default
[WARNING] Failed to launch x86-64-v3 version, staying with default
24384279 reads; of these:
  24384279 (100.00%) were unpaired; of these:
    1544443 (6.33%) aligned 0 times
    22025182 (90.33%) aligned exactly 1 time
    814654 (3.34%) aligned >1 times
93.67% overall alignment rate
[bam_sort_core] merging from 4 files and 4 in-memory blocks...
```

MACS2 peak calling:
For sample C:

```
pixi run macs3 callpeak \
  -t C_CHR1_sorted.bam \
  -c C_INR1_sorted.bam \
  -f BAM \
  -g 4.6e6 \
  --outdir SampleC \
  -n sorted_C \
  -B \
  --nomodel \
  --extsize 150 \
  -p 0.001
```

```
C_INR1_sorted.bam      D_INR1_sorted.bam.bai          E_coli_BW25113_index.2.bt2  pixi.lock                      SO_4933_C_CHR1_R1.fastq.gz     SO_4933_D_CHR1_R1.fastq.gz
C_INR1_sorted.bam.bai  E.coli_BW25113_annotation.bed  E_coli_BW25113_index.3.bt2  pixi.toml                      SO_4933_C_CHR1_R1_raw.bam      SO_4933_D_INR1_R1_bowtie2.log
D_CHR1_sorted.bam      E.coli_BW25113_annotation.gff  E_coli_BW25113_index.4.bt2  SampleC                        SO_4933_C_INR1_R1_bowtie2.log  SO_4933_D_INR1_R1.fastq.gz
(fastq_files_24_09) (base) ibab@IBAB-MSc-Comp11:~/bio-env/fastq_files_24_09$ pixi run macs3 callpeak \
  -t C_CHR1_sorted.bam \
  -c C_INR1_sorted.bam \
  -f BAM \
  -g 4.6e6 \
  --outdir SampleC \
  -n sorted_C \
  -B \
  -p 0.001
INFO  @ 24 Sep 2026 23:06:01: [33 MB]
# Command line: callpeak -t C_CHR1_sorted.bam -c C_INR1_sorted.bam -f BAM -g 4.6e6 --outdir SampleC -n sorted_C -B -p 0.001
# ARGUMENTS LIST:
# name = sorted_C
# format = BAM
# ChIP-seq file = ['C_CHR1_sorted.bam']
# control file = ['C_INR1_sorted.bam']
# effective genome size = 4.60e+06
# band width = 300
# model fold = [5, 50]
# pvalue cutoff = 1.00e-03
# qvalue will not be calculated and reported as -1 in the final output.
# The maximum gap between significant sites is assigned as the read length/tag size.
# The minimum length of peaks is assigned as the predicted fragment length "d".
# Larger dataset will be scaled towards smaller dataset.
# Range for calculating regional lambda is: 1000 bps and 10000 bps
# Broad region calling is off
# Paired-End mode is off

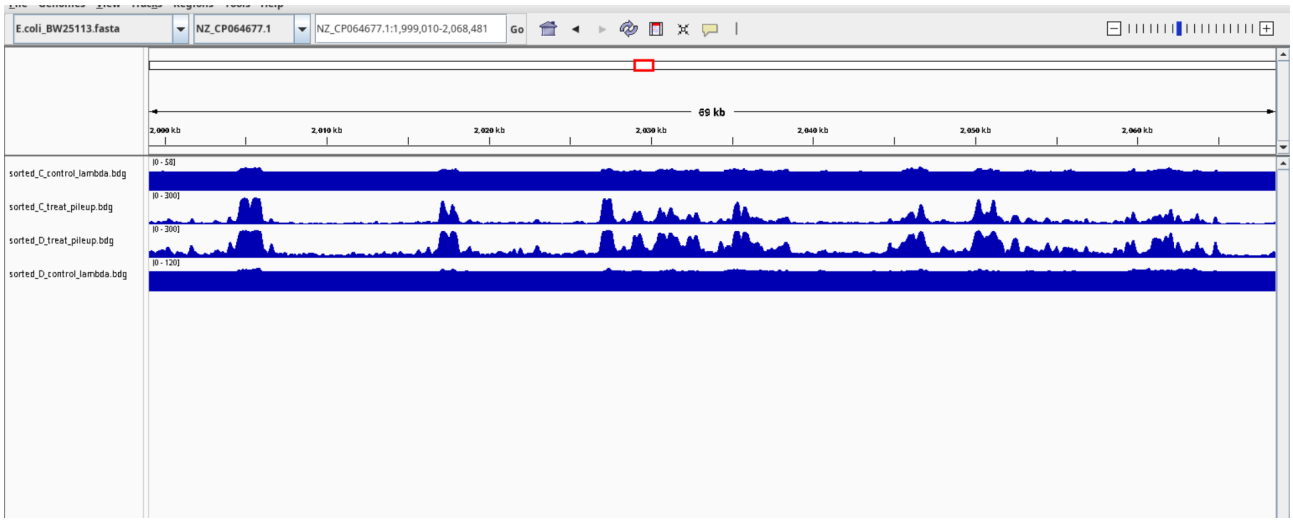
INFO  @ 24 Sep 2026 23:06:01: [33 MB] #1 read tag files...
INFO  @ 24 Sep 2026 23:06:01: [33 MB] #1 read treatment tags...
INFO  @ 24 Sep 2026 23:06:02: [62 MB]  1000000 reads parsed
```

For sample D:

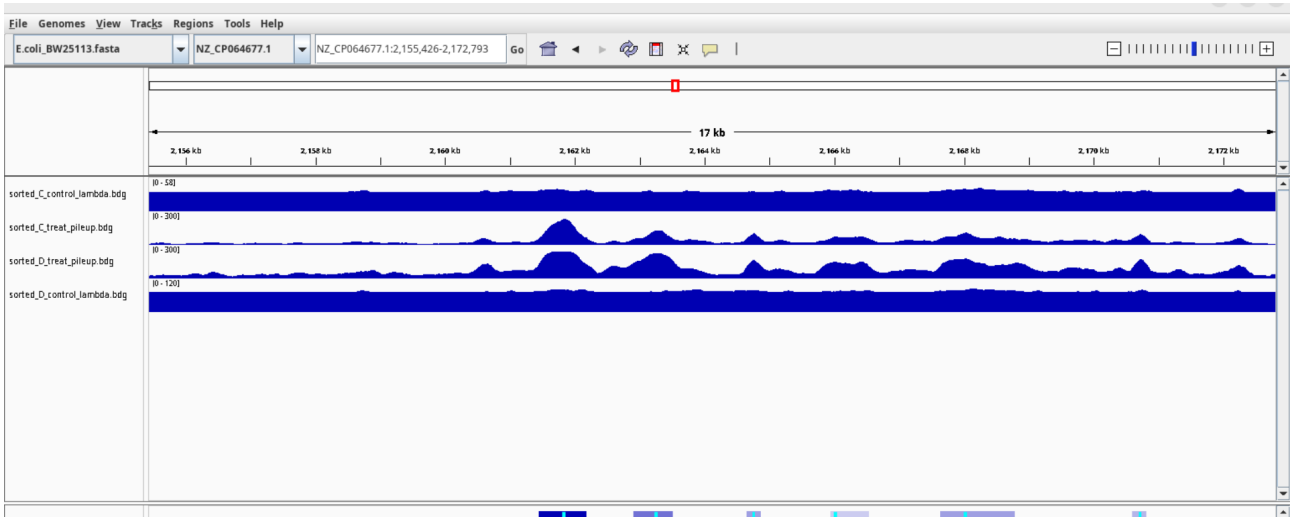
```
pixi run macs3 callpeak \
  -t D_CHR1_sorted.bam \
  -c D_INR1_sorted.bam \
  -f BAM \
  -g 4.6e6 \
  --outdir SampleD \
  -n sorted_D \
  -B \
  --nomodel \
  --extsize 150 \
  -p 0.001
```

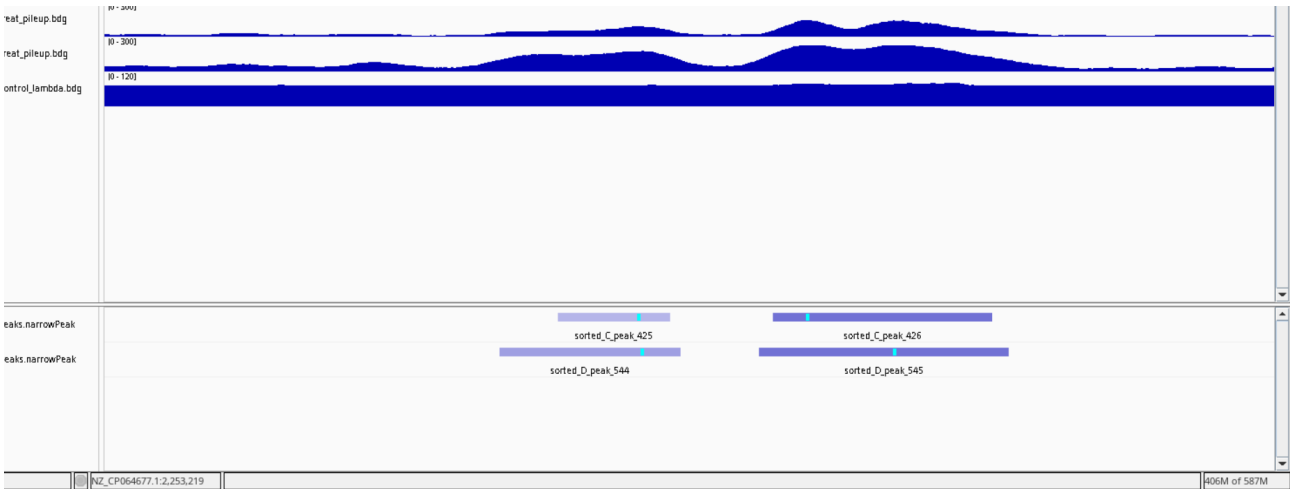
IGV:

```
cd ~/bio-env
pixi run igv
```



Between this two samples , both of them have almost same no of peaks, because they are from same transcription factors but sample C has less expression and sample D has more expression, so we can see some broader peaks in sample D and some more very less intensity peaks can be seen in the sample D but not in sample D;





Here , we can see that the peaks in the are broader in sample D , when we uploaded the narrow peak files in IGV(because the protein conc in sample D is higher)

Merging peaks:

pixi run mergePeaks ./SampleC/sorted_C_peaks.narrowPeak -d 300 > merge300.txt

pixi run mergePeaks ./SampleD/sorted_D_peaks.narrowPeak -d 300 > merge300_D.txt

```
INFO [Sep 24,2026 23:49] [GenomeManager] Loading genome: /home/ibab/bio-env/fastq_files_24_09/E.coli_BW25113.fasta
INFO [Sep 24,2026 23:49] [FastaUtils] Creating index file at /home/ibab/bio-env/fastq_files_24_09/E.coli_BW25113.fasta.fai
INFO [Sep 24,2026 23:49] [TrackLoader] Loading resource:  /home/ibab/bio-env/fastq_files_24_09/SampleC/sorted_C_peaks.narrowPeak
INFO [Sep 24,2026 23:51] [TrackLoader] Loading resource:  /home/ibab/bio-env/fastq_files_24_09/SampleC/sorted_C_control_lambda.bdg
INFO [Sep 24,2026 23:52] [TrackLoader] Loading resource:  /home/ibab/bio-env/fastq_files_24_09/SampleC/sorted_C_treat_pileup.bdg
INFO [Sep 24,2026 23:53] [TrackLoader] Loading resource:  /home/ibab/bio-env/fastq_files_24_09/SampleD/sorted_D_treat_pileup.bdg
INFO [Sep 24,2026 23:53] [TrackLoader] Loading resource:  /home/ibab/bio-env/fastq_files_24_09/SampleD/sorted_D_peaks.narrowPeak
INFO [Sep 24,2026 23:54] [TrackLoader] Loading resource:  /home/ibab/bio-env/fastq_files_24_09/SampleD/sorted_D_control_lambda.bdg
INFO [Sep 25,2026 00:00] [TrackLoader] Loading resource:  /home/ibab/bio-env/fastq_files_24_09/SampleC/sorted_C_peaks.narrowPeak
INFO [Sep 25,2026 00:00] [TrackLoader] Loading resource:  /home/ibab/bio-env/fastq_files_24_09/SampleD/sorted_D_peaks.narrowPeak
^CINFO [Sep 25,2026 00:03] [ShutdownThread] Shutting down
(fastq_files_24_09) (base) ibab@IBAB-MSc-Comp11:~/bio-env$ cd ~/bio-env/fastq_files_24_09

pixi run mergePeaks ./SampleC/sorted_C_peaks.narrowPeak -d 500 > merge500.txt
        Max distance to merge: 500 bp
        Merging single peak file...
        3.24% of 864 peaks were within 500 bp and merged
(fastq_files_24_09) (base) ibab@IBAB-MSc-Comp11:~/bio-env/fastq_files_24_09$ pixi run mergePeaks ./SampleC/sorted_C_peaks.narrowPeak -d 300 > merge300.txt
        Max distance to merge: 300 bp
        Merging single peak file...
        0.23% of 864 peaks were within 300 bp and merged
(fastq_files_24_09) (base) ibab@IBAB-MSc-Comp11:~/bio-env/fastq_files_24_09$
pixi run mergePeaks ./SampleD/sorted_D_peaks.narrowPeak -d 500 > merge500_D.txt^C
(fastq_files_24_09) (base) ibab@IBAB-MSc-Comp11:~/bio-env/fastq_files_24_09$ cd ~/bio-env/fastq_files_24_09

pixi run mergePeaks ./SampleD/sorted_D_peaks.narrowPeak -d 300 > merge300_D.txt
        Max distance to merge: 300 bp
        Merging single peak file...
        0.09% of 1082 peaks were within 300 bp and merged
(fastq_files_24_09) (base) ibab@IBAB-MSc-Comp11:~/bio-env/fastq_files_24_09$
```

Annotation:

for sample C:
pixi run annotatePeaks.pl ./SampleC/sorted_C_peaks.narrowPeak E.coli_BW25113.fasta -gff E.coli_BW25113_annotation.gff > annotated_peaks.tsv

annotated_peaks.tsv:

```
gion Size       Annotation      Detailed Annotation     Distance to TSS Nearest PromoterID      Entrez ID       Nearest Unigene Nearest Refseq  Nearest Ensembl Gene Name       Gene Alias   G
ene Description Gene Type
sorted_C_peak_451       NZ_CP064677.1   2456222 2457853 +       1344    NA      Intergenic      Intergenic      -2751   ID=exon-IVL04_RS11980-1;Parent=rna-IVL04_RS11980;anticodon=(pos:245982
1..2459823);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS11980;product=tRNA-Arg
sorted_C_peak_195       NZ_CP064677.1   1191858 1195120 +       1324    NA      Intergenic      Intergenic      88808   ID=exon-IVL04_RS06255-1;Parent=rna-IVL04_RS06255;Dbxref=RFAM:RF00391;N
ote=rtT sRNA%2C processed from tyrT transcript;gbkey=ncRNA;inference=COORDINATES: profile:INFERNAL:1.1.1;locus_tag=IVL04_RS06255;product=RtT sRNA
sorted_C_peak_401       NZ_CP064677.1   2130562 2131418 +       1315    NA      Intergenic      Intergenic      75249   ID=exon-IVL04_RS10155-1;Parent=rna-IVL04_RS10155;anticodon=(pos:205577
4..2055776);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS10155;product=tRNA-Asn
sorted_C_peak_252       NZ_CP064677.1   1459236 1459754 +       1313    NA      Intergenic      Intergenic      -176417 ID=exon-IVL04_RS06280-1;Parent=rna-IVL04_RS06280;anticodon=(pos:comple
ment(1283042..1283044));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS06280;product=tRNA-Tyr
sorted_C_peak_372       NZ_CP064677.1   2004508 2006048 +       1289    NA      Intergenic      Intergenic      -19680  ID=exon-IVL04_RS09790-1;Parent=rna-IVL04_RS09790;anticodon=(pos:comple
ment(1985563..1985565));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS09790;product=tRNA-Gly
sorted_C_peak_269       NZ_CP064677.1   1537831 1540603 +       1287    NA      Intergenic      Intergenic      -201475 ID=exon-IVL04_RS08545-1;Parent=rna-IVL04_RS08545;anticodon=(pos:174072
6..1740728);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS08545;product=tRNA-Val
sorted_C_peak_468       NZ_CP064677.1   2486458 2487592 +       1283    NA      Intergenic      Intergenic      24570   ID=exon-IVL04_RS12230-1;Parent=rna-IVL04_RS12230;anticodon=(pos:comple
ment(2511560..2511562));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS12230;product=tRNA-Ala
sorted_C_peak_75        NZ_CP064677.1   385193  386489  +       1281    NA      Intergenic      Intergenic      -86071  ID=exon-IVL04_RS02310-1;Parent=rna-IVL04_RS02310;Dbxref=RFAM:RF00169;g
bkey=ncRNA;gene=ffs;inference=COORDINATES: profile:INFERNAL:1.1.1;locus_tag=IVL04_RS02310;product=signal recognition particle sRNA small type
sorted_C_peak_364       NZ_CP064677.1   1951865 1952728 +       1279    NA      Intergenic      Intergenic      33086   ID=exon-IVL04_RS09780-1;Parent=rna-IVL04_RS09780;anticodon=(pos:comple
ment(1985346..1985348));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS09780;product=tRNA-Leu
sorted_C_peak_268       NZ_CP064677.1   1523971 1527408 +       1279    NA      Intergenic      Intergenic      -215003 ID=exon-IVL04_RS08545-1;Parent=rna-IVL04_RS08545;anticodon=(pos:174072
6..1740728);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS08545;product=tRNA-Val
sorted_C_peak_198       NZ_CP064677.1   1205586 1208652 +       1277    NA      Intergenic      Intergenic      75178   ID=exon-IVL04_RS06255-1;Parent=rna-IVL04_RS06255;Dbxref=RFAM:RF00391;N
ote=rtT sRNA%2C processed from tyrT transcript;gbkey=ncRNA;inference=COORDINATES: profile:INFERNAL:1.1.1;locus_tag=IVL04_RS06255;product=RtT sRNA
sorted_C_peak_102       NZ_CP064677.1   553282  555990  +       1276    NA      Intergenic      Intergenic      -5543   ID=exon-IVL04_RS02710-1;Parent=rna-IVL04_RS02710;anticodon=(pos:560213
..560215);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS02710;product=tRNA-Arg
sorted_C_peak_142       NZ_CP064677.1   728680  729653  +       1276    NA      Intergenic      Intergenic      -36577  ID=exon-IVL04_RS03405-1;Parent=rna-IVL04_RS03405;anticodon=(pos:comple
ment(692553..692555));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS03405;product=tRNA-Met
sorted_C_peak_350       NZ_CP064677.1   1864304 1865896 +       1273    NA      Intergenic      Intergenic      100471  ID=exon-IVL04_RS08665-1;Parent=rna-IVL04_RS08665;Dbxref=RFAM:RF00034;g
bkey=ncRNA;gene=rprA;inference=COORDINATES: profile:INFERNAL:1.1.1;locus_tag=IVL04_RS08665;product=antisense sRNA RprA
sorted_C_peak_139       NZ_CP064677.1   715604  716625  +       1271    NA      Intergenic      Intergenic      -23525  ID=exon-IVL04_RS03405-1;Parent=rna-IVL04_RS03405;anticodon=(pos:comple
ment(692553..692555));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS03405;product=tRNA-Met
sorted_C_peak_403       NZ_CP064677.1   2161447 2162186 +       1271    NA      Intergenic      Intergenic      106075  ID=exon-IVL04_RS10155-1;Parent=rna-IVL04_RS10155;anticodon=(pos:205577
4..2055776);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS10155;product=tRNA-Asn
sorted_C_peak_91        NZ_CP064677.1   522424  523470  +       1266    NA      Intergenic      Intergenic      -37232  ID=exon-IVL04_RS02710-1;Parent=rna-IVL04_RS02710;anticodon=(pos:560213
..560215);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS02710;product=tRNA-Arg
sorted_C_peak_5 NZ_CP064677.1   41732   42547   +       1265    NA      Intergenic      Intergenic      -178119 ID=exon-IVL04_RS01000-1;Parent=rna-IVL04_RS01000;Dbxref=RFAM:RF00177;gbkey=rRN
A;inference=COORDINATES: profile:INFERNAL:1.1.1;locus_tag=IVL04_RS01000;product=16S ribosomal RNA
sorted_C_peak_618       NZ_CP064677.1   3405888 3407387 +       1262    NA      Intergenic      Intergenic      10262   ID=exon-IVL04_RS16570-1;Parent=rna-IVL04_RS16570;Dbxref=RFAM:RF00001;g
bkey=rRNA;gene=rrf;inference=COORDINATES: profile:INFERNAL:1.1.1;locus_tag=IVL04_RS16570;product=5S ribosomal RNA
```

We can see that most of the regions are intergenic regions , for this case as the proteins are trasncription factor so these could be promoter regions

For sample D:
pixi run annotatePeaks.pl ./SampleD/sorted_D_peaks.narrowPeak E.coli_BW25113.fasta -gff E.coli_BW25113_annotation.gff > annotated_peaks_D.tsv

```
gion Size       Annotation      Detailed Annotation     Distance to TSS Nearest PromoterID      Entrez ID       Nearest Unigene Nearest Refseq  Nearest Ensembl Gene Name       Gene Alias   G
ene Description Gene Type
sorted_D_peak_385       NZ_CP064677.1   1630868 1633444 +       691     NA      Intergenic      Intergenic      -108536 ID=exon-IVL04_RS08545-1;Parent=rna-IVL04_RS08545;anticodon=(pos:174072
6..1740728);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS08545;product=tRNA-Val
sorted_D_peak_330       NZ_CP064677.1   1430702 1431591 +       689     NA      Intergenic      Intergenic      -148068 ID=exon-IVL04_RS06280-1;Parent=rna-IVL04_RS06280;anticodon=(pos:comple
ment(1283042..1283044));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS06280;product=tRNA-Tyr
sorted_D_peak_561       NZ_CP064677.1   2357667 2359507 +       684     NA      Intergenic      Intergenic      78897   ID=exon-IVL04_RS11185-1;Parent=rna-IVL04_RS11185;anticodon=(pos:227972
4..2279726);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS11185;product=tRNA-Pro
sorted_D_peak_465       NZ_CP064677.1   1940197 1941068 +       675     NA      Intergenic      Intergenic      44750   ID=exon-IVL04_RS09780-1;Parent=rna-IVL04_RS09780;anticodon=(pos:comple
ment(1985346..1985348));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS09780;product=tRNA-Leu
sorted_D_peak_359       NZ_CP064677.1   1550431 1550947 +       673     NA      Intergenic      Intergenic      -190003 ID=exon-IVL04_RS08545-1;Parent=rna-IVL04_RS08545;anticodon=(pos:174072
6..1740728);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS08545;product=tRNA-Val
sorted_D_peak_512       NZ_CP064677.1   2121974 2124184 +       671     NA      Intergenic      Intergenic      67338   ID=exon-IVL04_RS10155-1;Parent=rna-IVL04_RS10155;anticodon=(pos:205577
4..2055776);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS10155;product=tRNA-Asn
sorted_D_peak_322       NZ_CP064677.1   1406749 1408423 +       668     NA      Intergenic      Intergenic      -124508 ID=exon-IVL04_RS06280-1;Parent=rna-IVL04_RS06280;anticodon=(pos:comple
ment(1283042..1283044));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS06280;product=tRNA-Tyr
sorted_D_peak_376       NZ_CP064677.1   1609898 1611043 +       668     NA      Intergenic      Intergenic      -130222 ID=exon-IVL04_RS08545-1;Parent=rna-IVL04_RS08545;anticodon=(pos:174072
6..1740728);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS08545;product=tRNA-Val
sorted_D_peak_325       NZ_CP064677.1   1417817 1419775 +       668     NA      Intergenic      Intergenic      -135718 ID=exon-IVL04_RS06280-1;Parent=rna-IVL04_RS06280;anticodon=(pos:comple
ment(1283042..1283044));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS06280;product=tRNA-Tyr
sorted_D_peak_355       NZ_CP064677.1   1520897 1522176 +       660     NA      Intergenic      Intergenic      -219156 ID=exon-IVL04_RS08545-1;Parent=rna-IVL04_RS08545;anticodon=(pos:174072
6..1740728);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS08545;product=tRNA-Val
sorted_D_peak_297       NZ_CP064677.1   1308809 1310783 +       660     NA      Intergenic      Intergenic      -26718  ID=exon-IVL04_RS06280-1;Parent=rna-IVL04_RS06280;anticodon=(pos:comple
ment(1283042..1283044));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS06280;product=tRNA-Tyr
sorted_D_peak_369       NZ_CP064677.1   1575700 1578663 +       659     NA      Intergenic      Intergenic      -163511 ID=exon-IVL04_RS08545-1;Parent=rna-IVL04_RS08545;anticodon=(pos:174072
6..1740728);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS08545;product=tRNA-Val
sorted_D_peak_450       NZ_CP064677.1   1864204 1866317 +       659     NA      Intergenic      Intergenic      100631  ID=exon-IVL04_RS08665-1;Parent=rna-IVL04_RS08665;Dbxref=RFAM:RF00034;g
bkey=ncRNA;gene=rprA;inference=COORDINATES: profile:INFERNAL:1.1.1;locus_tag=IVL04_RS08665;product=antisense sRNA RprA
sorted_D_peak_427       NZ_CP064677.1   1781292 1781856 +       657     NA      Intergenic      Intergenic      16945   ID=exon-IVL04_RS08665-1;Parent=rna-IVL04_RS08665;Dbxref=RFAM:RF00034;g
bkey=ncRNA;gene=rprA;inference=COORDINATES: profile:INFERNAL:1.1.1;locus_tag=IVL04_RS08665;product=antisense sRNA RprA
sorted_D_peak_532       NZ_CP064677.1   2197464 2198239 +       655     NA      Intergenic      Intergenic      -81839  ID=exon-IVL04_RS11185-1;Parent=rna-IVL04_RS11185;anticodon=(pos:227972
4..2279726);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS11185;product=tRNA-Pro
sorted_D_peak_386       NZ_CP064677.1   1634763 1636696 +       652     NA      Intergenic      Intergenic      -104963 ID=exon-IVL04_RS08545-1;Parent=rna-IVL04_RS08545;anticodon=(pos:174072
6..1740728);gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS08545;product=tRNA-Val
sorted_D_peak_429       NZ_CP064677.1   1796814 1799460 +       650     NA      Intergenic      Intergenic      33508   ID=exon-IVL04_RS08665-1;Parent=rna-IVL04_RS08665;Dbxref=RFAM:RF00034;g
bkey=ncRNA;gene=rprA;inference=COORDINATES: profile:INFERNAL:1.1.1;locus_tag=IVL04_RS08665;product=antisense sRNA RprA
sorted_D_peak_459       NZ_CP064677.1   1901582 1902635 +       650     NA      Intergenic      Intergenic      83274   ID=exon-IVL04_RS09780-1;Parent=rna-IVL04_RS09780;anticodon=(pos:comple
ment(1985346..1985348));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS09780;product=tRNA-Leu
sorted_D_peak_662       NZ_CP064677.1   2852902 2853863 +       648     NA      Intergenic      Intergenic      -41378  ID=exon-IVL04_RS13715-1;Parent=rna-IVL04_RS13715;anticodon=(pos:comple
ment(2811968..2811970));gbkey=tRNA;inference=COORDINATES: profile:tRNAscan-SE:2.0.7;locus_tag=IVL04_RS13715;product=tRNA-Ser
```

Identifying closest genes: (by Bedtools)-so we can look for eg what are the genes that are closer to the peaks, we have to give the distance for eg 1000 bp both sides around the peak

pixi run bedtools closest -a ./SampleC/sorted_C_peaks.narrowPeak -b E.coli_BW25113_annotation.bed | awk -F "\t" '{if($18 == "gene"){print}}' > C_closest_genes.bed

pixi run bedtools closest -a ./SampleD/sorted_D_peaks.narrowPeak -b E.coli_BW25113_annotation.bed | awk -F "\t" '{if($18 == "gene"){print}}' > D_closest_genes.bed

```
1 NZ_CP064677.1   16987   17538   sorted_C_peak_1 723     .       4.42603 72.3783 70.8833 176     NZ_CP064677.1   17488   18655   gene-
  IVL04_RS00085   .       +       RefSeq  gene    .       ID=gene-
  IVL04_RS00085;Name=nhaA;gbkey=Gene;gene=nhaA;gene_biotype=protein_coding;locus_tag=IVL04_RS00085;old_locus_tag=IVL04_00085
2 NZ_CP064677.1   33966   34537   sorted_C_peak_3 464     .       3.66443 46.4996 45.1711 166     NZ_CP064677.1   30816   34038   gene-
  IVL04_RS00155   .       +       RefSeq  gene    .       ID=gene-
  IVL04_RS00155;Name=carB;gbkey=Gene;gene=carB;gene_biotype=protein_coding;locus_tag=IVL04_RS00155;old_locus_tag=IVL04_00155
3 NZ_CP064677.1   33966   34537   sorted_C_peak_3 464     .       3.66443 46.4996 45.1711 166     NZ_CP064677.1   34299   34695   gene-
  IVL04_RS00160   .       +       RefSeq  gene    .       ID=gene-
  IVL04_RS00160;Name=caiF;gbkey=Gene;gene=caiF;gene_biotype=protein_coding;locus_tag=IVL04_RS00160;old_locus_tag=IVL04_00160
4 NZ_CP064677.1   36004   36558   sorted_C_peak_4 658     .       4.18839 65.8057 64.3552 272     NZ_CP064677.1   35376   36162   gene-
  IVL04_RS00170   .       -       RefSeq  gene    .       ID=gene-
  IVL04_RS00170;Name=caiD;gbkey=Gene;gene=caiD;gene_biotype=protein_coding;locus_tag=IVL04_RS00170;old_locus_tag=IVL04_00170
5 NZ_CP064677.1   36004   36558   sorted_C_peak_4 658     .       4.18839 65.8057 64.3552 272     NZ_CP064677.1   36270   37824   gene-
  IVL04_RS00175   .       -       RefSeq  gene    .       ID=gene-
  IVL04_RS00175;Name=caiC;gbkey=Gene;gene=caiC;gene_biotype=protein_coding;locus_tag=IVL04_RS00175;old_locus_tag=IVL04_00175
6 NZ_CP064677.1   41731   42547   sorted_C_peak_5 1265    .       5.90796 126.501 123.075 250     NZ_CP064677.1   40416   41931   gene-
  IVL04_RS00190   .       -       RefSeq  gene    .       ID=gene-
  IVL04_RS00190;Name=caiT;gbkey=Gene;gene=caiT;gene_biotype=protein_coding;locus_tag=IVL04_RS00190;old_locus_tag=IVL04_00190
7 NZ_CP064677.1   41731   42547   sorted_C_peak_5 1265    .       5.90796 126.501 123.075 250     NZ_CP064677.1   42402   43173   gene-
  IVL04_RS00195   .       +       RefSeq  gene    .       ID=gene-
  IVL04_RS00195;Name=fixA;gbkey=Gene;gene=fixA;gene_biotype=protein_coding;locus_tag=IVL04_RS00195;old_locus_tag=IVL04_00195
8 NZ_CP064677.1   58284   58862   sorted_C_peak_6 363     .       3.32224 36.3563 35.0921 113     NZ_CP064677.1   58473   59124   gene-
  IVL04_RS00270   .       +       RefSeq  gene    .       ID=gene-
  IVL04_RS00270;Name=IVL04_RS00270;gbkey=Gene;gene_biotype=protein_coding;locus_tag=IVL04_RS00270;old_locus_tag=IVL04_00270
9 NZ_CP064677.1   59134   59293   sorted_C_peak_7 140     .       2.28809 14.0504 12.9604 95      NZ_CP064677.1   59120   59279   gene-
  IVL04_RS00275   .       +       RefSeq  gene    .       ID=gene-
  IVL04_RS00275;Name=IVL04_RS00275;gbkey=Gene;gene_biotype=protein_coding;locus_tag=IVL04_RS00275;old_locus_tag=IVL04_00275
10 NZ_CP064677.1  66502   67876   sorted_C_peak_8 1121    .       5.14588 112.197 110.27  463     NZ_CP064677.1   65854   66574   gene-
```

Step 2: Extracting Peak DNA Sequences (`bedtools getfasta`)

First, ensured the FASTA file has an index (`.fai`):

pixi run samtools faidx E.coli_BW25113.fasta

For C:

pixi run bedtools **getfasta** -fi E.coli_BW25113.fasta -bed ./SampleC/sorted_C_peaks.narrowPeak -fo C_seq.fasta

for D:

pixi run bedtools **getfasta** -fi E.coli_BW25113.fasta -bed ./SampleD/sorted_D_peaks.narrowPeak -fo D_seq.fasta

bedtools' **getfasta :** It returns a multifasta file , for every peak it will return one FASTA deq,,,so for eg if we have 990 peaks we will have 990 fasta sequnces;

Step 3: Motif Discovery using MEME

For Sample C:

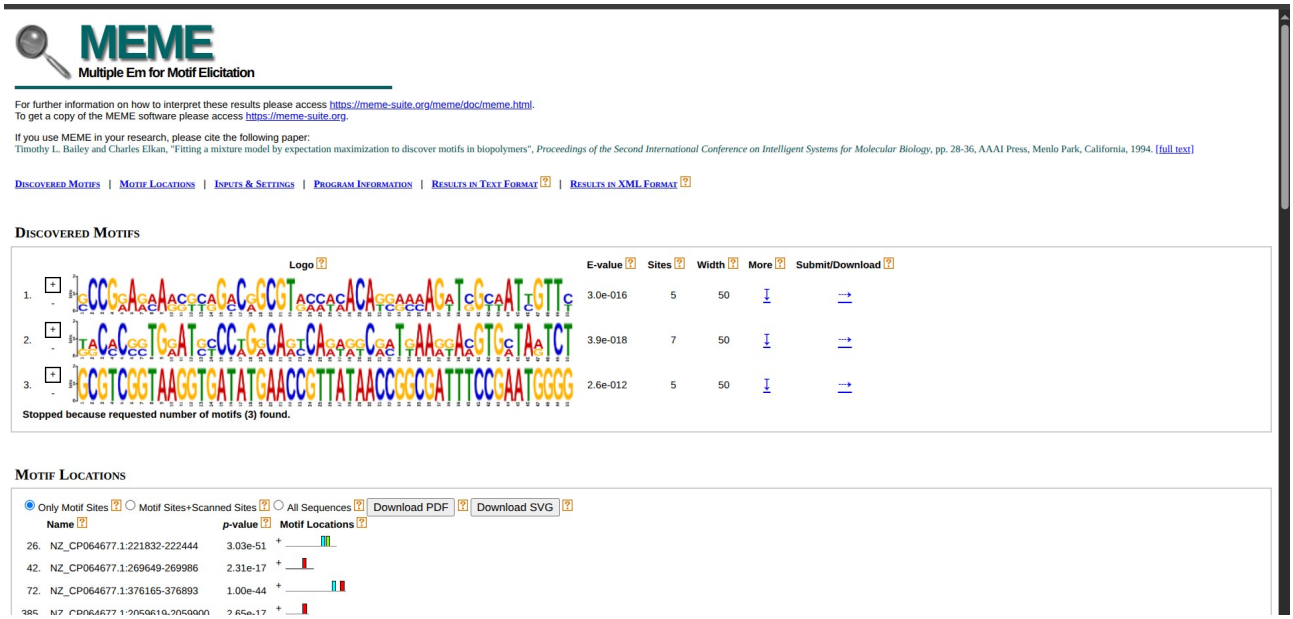
Bash

```
pixi run meme C_seq.fasta -dna -nmotifs 3 -oc meme_out_C
```

For Sample D:

Bash

```
pixi run meme D_seq.fasta -dna -nmotifs 3 -oc meme_out_D
```



Here the TF does not shows a motif specific bindng , what we get -is a long stretch of motifs, but we can see mostly they are AT rich,,so this is the only specificity of this TF

Discovered Motifs

	Logo	E-value	Sites	Width	More	Submit/Download
1.		3.0e-016	5	50		
2.		3.9e-018	7	50		
3.		2.6e-012	5	50		

Stopped because requested number of motifs (3) found.